

SATURN

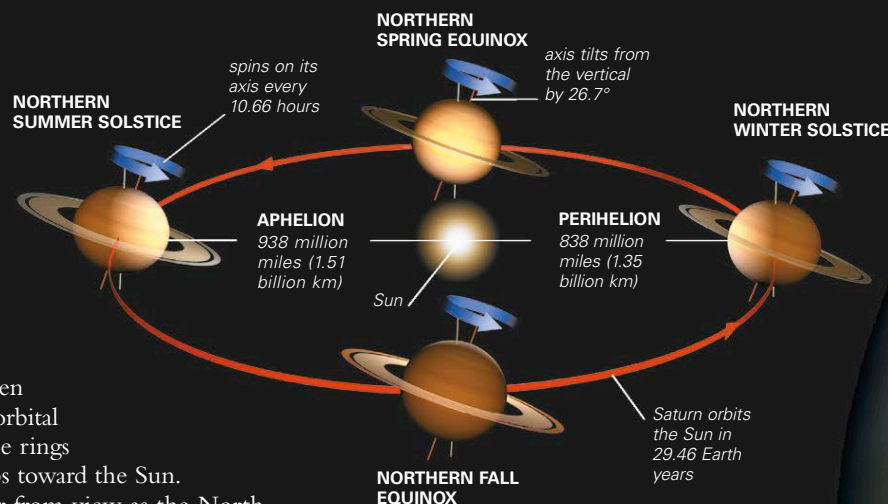
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SATURN IS THE SECOND-LARGEST PLANET and the sixth from the Sun—it is the most distant planet normally visible to the naked eye. A huge ball of gas and liquid, Saturn has a bulging equator and an internal energy source. With a composition dominated by hydrogen, it is the least dense of all the planets. A spectacular system of rings encircles the planet itself, and it also has a large family of moons.

ORBIT

Saturn takes 29.46 Earth years to complete one orbit of the Sun. It is tilted to its orbital plane by 26.7° , a little more than Earth's axial tilt. This means that as Saturn moves along its orbit, the north and south poles take turns pointing toward the Sun. The changing orientation of Saturn to the Sun is seen from Earth by the apparent opening and closing of the planet's ring system. The rings are seen edge-on, for example, at the start of an orbital period. Then an increasing portion of the rings is seen from above as the North Pole tips toward the Sun.

The rings slowly close up and disappear from view as the North Pole starts to tip away until, 14.73 Earth years (half an orbit) later, they appear edge-on again. Now the South Pole tips sunward and the rings are seen increasingly from below. They close up once again as the South Pole turns away, until they are seen edge-on once more as the orbit is completed. The strength of the Sun at Saturn is only about 1 percent of that received on Earth, but it is enough to generate seasonal smog. Saturn is at perihelion at the time the South Pole is facing the Sun.

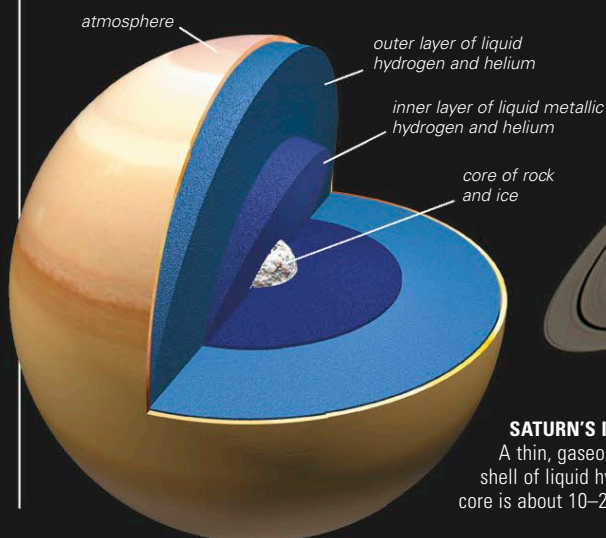


SPIN AND ORBIT

Saturn spins on its axis as it orbits. The rapid spin flings material outward, with the result that Saturn is about 10 percent wider at its equator than its poles. Its bulging equator is bigger than that of any other planet.

STRUCTURE

Saturn's mass is only 95 times that of Earth, yet 764 Earths could fit inside it. This is because Saturn is composed mainly of the lightest elements, hydrogen and helium, which are in both gaseous and liquid states. Saturn is the least dense of all the planets. If it were possible to put Saturn in an ocean of water, it would float. The planet has no discernible surface—its outer layer is gaseous atmosphere. Inside the planet, pressure and temperature increase with depth and the hydrogen and helium molecules are forced closer and closer together until they become fluid. Deeper still, the atoms are stripped of their electrons and act as a liquid metal. Electric currents within this region generate a magnetic field 71 percent the strength of Earth's (see p.127).



SATURN'S INTERIOR

A thin, gaseous atmosphere surrounds a vast shell of liquid hydrogen and helium. The central core is about 10–20 times the mass of Earth.

RINGLEADER

Girdled by its bright system of rings, Saturn has a hazy, muted appearance in this Cassini image, which shows the planet in its natural colors. A number of Saturn's moons are also visible in this image.